Lesson 11 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

This Is a Milestone Lesson

Once multiple waves work today, the capstone reaches the Level 2 Complete milestone from the Feature-to-Lesson Map - the second of the year's big checkpoints, after Lesson 6's Level 1 MVP. This deserves the same extra celebratory framing Lesson 6 got: a dedicated Milestone Moment discussion, and a real share-out at the end of lab, not just a quick check.

- ☐ Read the Teacher's Script's Milestone Moment section before class, and plan to give it genuine time - this is a good one to slow down for.

This Week: One Question Becomes Two

Since Lesson 6 the win check has asked one question: 'are all the enemies gone?' This week it grows a second: 'AND was that the last wave?' If yes to both, it is a real win. If the enemies are gone but more waves remain, the round continues - a fresh set spawns, and score, lives and game_active are untouched.

The routine that creates a set of enemies already exists from Lesson 10 (spawn_enemies in Godot, GameMaker and Scratch, SpawnEnemies in Unity). It is now called from a third place: not at launch, not on restart, but by the game itself when a wave is cleared. Every new piece of shared state (wave, total_waves) lives on the GameManager, per Rule 13.

- ☐ Read each engine's Handout once before class. Every engine restructures its existing win check into a nested if/else this week.

Game State, Again

Lesson 10 taught students to audit game state on restart. This week adds new state (wave, total_waves, and for Level 3 an on-screen wave message). Watch for the classic miss: wave not reset on restart, so a game restarted from wave 3 starts at wave 3.

- ☐ In your own test project, restart from wave 2 and from wave 3 (after a win and after a loss) and confirm the HUD shows wave 1.

The Stalemate Is Back - On Purpose

The Stalemate bug from Lesson 10 (an escaped enemy lowers lives but never lowers enemies_remaining, so a round with some kills and some escapes ends with no win, no loss and no restart) is deliberately NOT fixed this week. Waves make it worse: it can now freeze the

game at any of the three waves. The goal is for it to become irritating enough that students go back and fix it themselves, so please do not hand them the fix.

- ☐ Students who fixed the Stalemate in Lesson 10 Level 3 should confirm their fix still works with waves. Each Key's Level 3 section has a worked, wave-aware version if you need to compare.
- ☐ If a student asks why a wave never ends after enemies escaped: name it (the Stalemate), confirm it is real and not a typo, and send them to Lesson 10 Level 3.

Self-Contained Lesson Assets

No new assets this week - waves reuse the exact same enemy spawn logic and placeholder sprites already in every project.

Continuity From Lesson 10

- ☐ Remind students to open their own [YourName]Game project from Lesson 10, not create a new one.
- ☐ If a student's Lesson 10 project is lost or broken, have a pre-built "Lesson 10 complete" starter available per engine (from the Lesson 10 Teacher's Keys) so they aren't blocked.

Engine Verification (active engines)

Only Scratch, Godot, GameMaker (script and Visual) and Unity (C#) are in active use this term. The other engines are paused until the book pass and have no Lesson 11 materials in this release.

- ☐ Test a full 3-wave playthrough on each active engine before class: clear wave 1 (round continues), clear wave 2 (round continues), clear wave 3 (real win, then R returns to wave 1).
- ☐ Confirm losing the last life mid-wave (at any wave number) still triggers Game Over, not a wave transition.
- ☐ Godot: confirm a wave transition spawns enemies with no errors in the Debugger. The next wave is spawned with call_deferred because the wave is cleared in the middle of a collision; if you see a 'flushing queries' error, that call is missing its deferral.
- ☐ Unity: confirm the HUD's new Wave Text slot is filled in the Inspector, and that waves spawn (the next wave is requested by a static flag and spawned from GameManager's Update()).
- ☐ Scratch: confirm each wave spawns exactly five enemies, never ten (a doubled wave usually means the Enemy sprite's spawn script is still on [start game] as well as [spawn enemies]).
- ☐ GameMaker Visual: confirm the else branch of the win check contains all three blocks and that spawn_enemies is called with a Function Call block, not Execute Script.

Materials

- ☐ Print or queue digital copies of the Student Handouts for the four active engines (or confirm students can access their engine's version electronically).
- ☐ Have the Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies.

Known Gotchas This Week

- The most likely mistake: forgetting the wave check entirely, so clearing wave 1 immediately triggers the full win. The win message must NOT appear until wave 3 is cleared.
- A close second: the wave-advance branch also resetting score or lives. Only enemies_remaining and wave should change between waves.
- Restarting mid-way through wave 2 or 3 must return to wave 1: test it directly, not just from a fresh start.
- This is a genuinely exciting lesson to finish - encourage a full 3-wave playthrough as part of the end-of-lesson share-out, with the same energy as Lesson 6's milestone celebration.